

SMART INDIA HACKATHON 2026



ORBITAL-HAR

Offline on-board procedure supervision for BAS experiments

Problem Statement ID: SIH26174

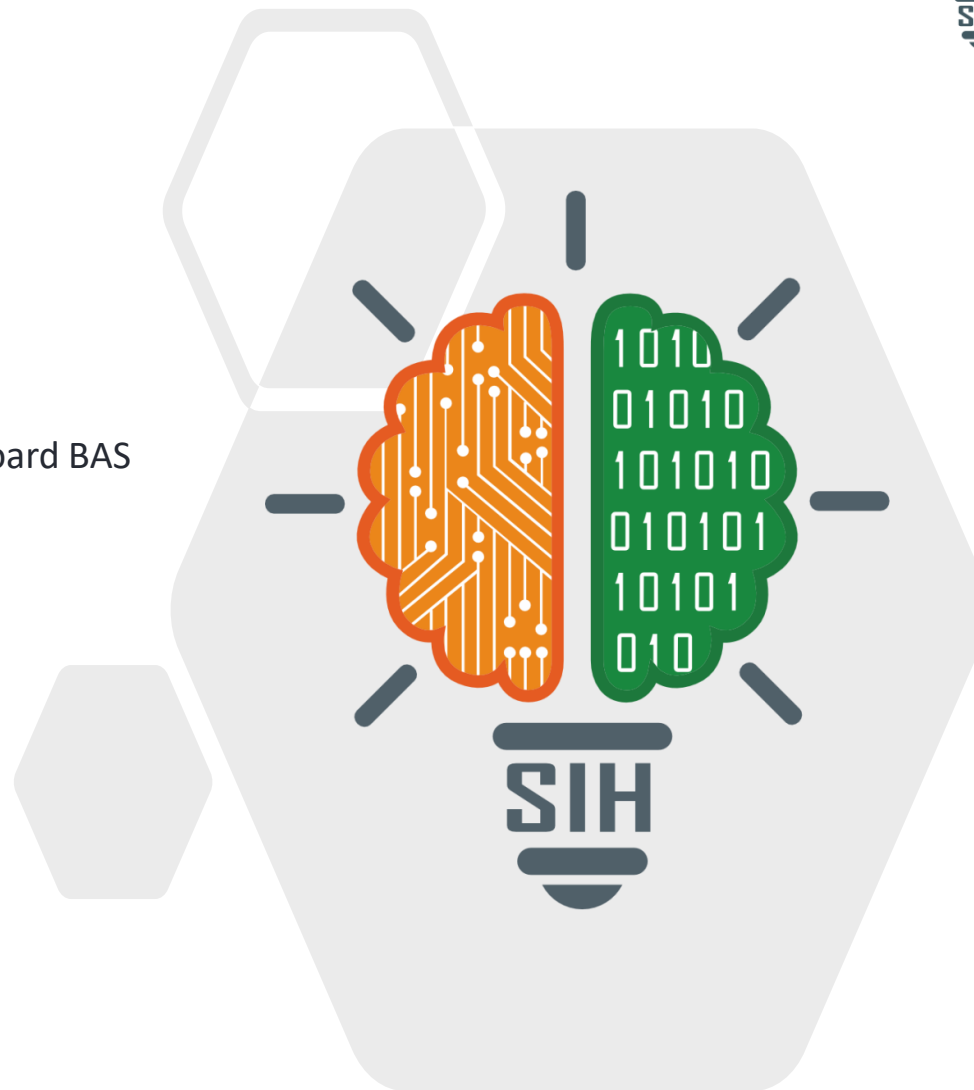
Problem Statement Title: AI Human Activity Recognition for On-board BAS Experiments

Theme: Space Technology

PS Category: Software

Team ID: (as registered on SIH portal)

Team Name: Hashira



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The idea in one line

A fixed payload camera watches an astronaut run an experiment. The system recognises each action, prompts the next step, voice-alerts on a skipped or out-of-order step, and emits a few KB of verified telemetry instead of a video downlink — fully offline on the edge.

How it addresses the problem

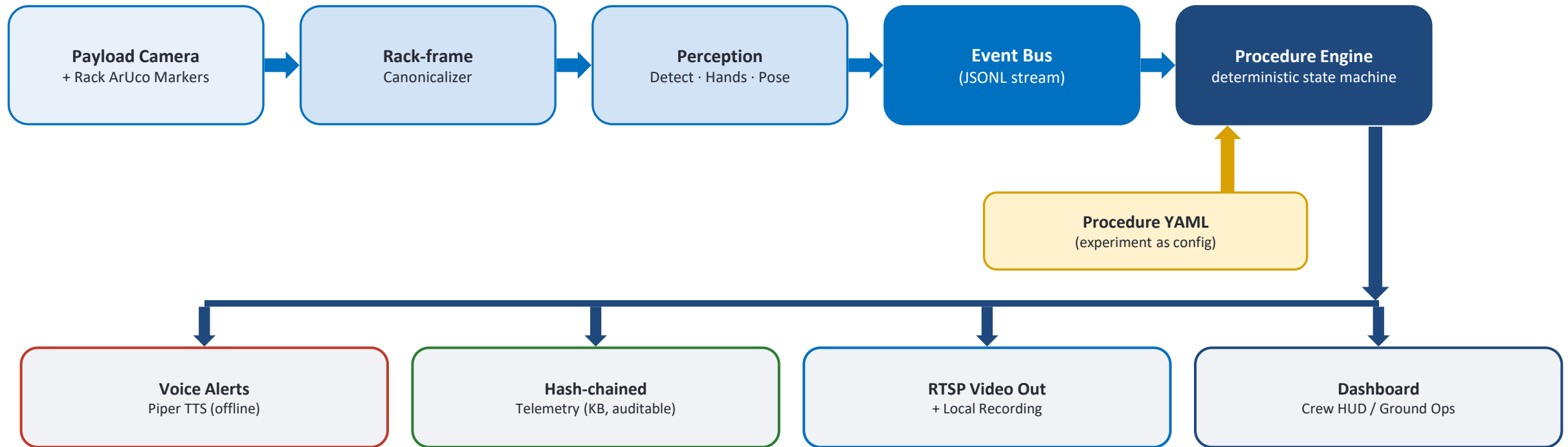
- Replaces high-bandwidth video downlink with tamper-evident, hash-chained step telemetry — ground control gets provable status without real-time comms.
- Reasoning-first, never guesses: perception publishes primitives (objects, hands, pose) to an event bus; a deterministic state machine decides and, below confidence, reports UNVERIFIED rather than a confident wrong verdict.
- Orientation-agnostic: the input frame is canonicalised to the payload rack (ArUco), so a floating or inverted astronaut is still recognised — solving the “no fixed up/down” problem.

Innovation & uniqueness

- Procedure-as-config: every experiment is a YAML file, not code — a new experiment is a new file, with zero changes to the engine.
- Event-bus seam decouples AI from reasoning: enables record-once/replay-1000 × tuning and “demo insurance” if a camera fails live.
- Explainable + auditable: deterministic verdicts and an append-only hash-chained log make every decision defensible.

Technologies

Python 3.11 · YOLO11 detection (object states as classes) · MediaPipe Hands · YOLO11-pose · OpenCV ArUco (rack frame) · Pydantic + YAML procedures · Piper offline TTS · FastAPI + WebSocket · React + Vite (bundled, no CDN) · FFmpeg + MediaMTX (RTSP + local record) · Blender synthetic data · Jetson Orin Nano + TensorRT (edge)



Fire-and-forget outputs — losing any output never stops supervision.

FEASIBILITY AND VIABILITY



Feasibility — already de-risked

- The reasoning spine is built and tested end-to-end with no camera and no models: procedure schema, event bus with replay, 7 predicate evaluators, step state machine and CLI — 73 tests pass, plus a golden replay corpus.
- A live webcam demo already runs today using a pretrained YOLO stand-in feeding the real engine (on-screen HUD + offline voice).
- Small-data regime: fixed camera, controlled scene, under 12 classes — ~2,000 synthetic + ~800 real frames is enough to fine-tune YOLO11 from COCO weights.

Challenges & risks

- Dataset generation is the critical path (props → footage → labels → model). Standard pose models fail on inverted bodies. Edge compute/power budget and real-time alert latency are constrained.

Strategies to overcome them

- Blender auto-labelled synthetic data randomised across the full orientation sphere; canonicalise the input frame (not the output pose); pre-synthesize voice prompts to WAV off the alert path; TensorRT export for the Jetson; event-bus replay as live demo insurance.

IMPACT AND BENEFITS



Mission impact

- Enables autonomous, protocol-perfect science execution beyond real-time ground support (BAS, lunar) — the core need in the problem statement.

Bandwidth saved

- KB-scale verified telemetry instead of GB-scale video downlink — orders-of-magnitude saving on a data-restricted space link.

Safety & trust

- Deterministic, explainable verdicts and a tamper-evident hash-chained record of every step reduce crew error and support post-mission audit.

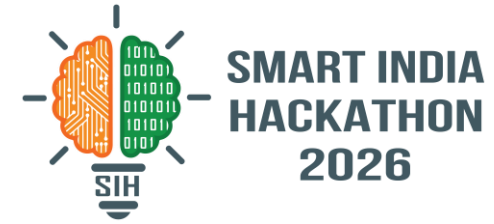
Reusability & spin-off

- Procedure-as-config makes it a platform, not a one-experiment demo — any procedural task (surgical, industrial, hazardous-site, training) reuses the same engine.

Economic & social

- Lowers ground-ops load and crew training burden; indigenous, offline edge-AI capability aligned with India's space programme.

RESEARCH AND REFERENCES



- SIH 2026 Problem Statement SIH26174 — AI HAR for On-board BAS Experiments, ISRO / Department of Space (sih.gov.in/sih2026PS)
- Ultralytics YOLO11 / YOLO-World — object detection & open-vocabulary pre-labelling (docs.ultralytics.com)
- Google MediaPipe Hands — real-time hand landmarks & hand-object interaction
- OpenCV ArUco — 6-DoF fiducial marker pose for rack-frame canonicalisation
- Piper TTS (Rhasspy) — fully offline neural text-to-speech for voice alerts
- Orientation-agnostic 3D Human Mesh Recovery (HMR) literature — rack-relative body tracking
- Blender bpy — scripted synthetic-data generation with automatic labels